

Experiment 24

Design a C program to demonstrate UNIX system calls for file management.

C Program

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <string.h>

int main() {
    int fd;
    char buffer[100];

    fd = open("sample.txt", O_CREAT | O_RDWR, 0644);
    if (fd < 0) {
        perror("open");
        return 1;
    }

    strcpy(buffer, "UNIX file management using system calls.\n");
    write(fd, buffer, strlen(buffer));

    lseek(fd, 0, SEEK_SET);

    int n = read(fd, buffer, sizeof(buffer) - 1);
    buffer[n] = '\0';

    printf("File contents: %s", buffer);

    close(fd);
    printf("File opened, written, read and closed successfully.\n");
    return 0;
}
```

Sample Output

```
File contents: UNIX file management using system calls.
File opened, written, read and closed successfully.
```

Result

UNIX file management system calls were successfully demonstrated using open, write, lseek, read and close.